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EXTENDED ABSTRACT

Biogeochemical Forecastability Decomposition via Progressive Information Scaling

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Received xx xxx xxxx

1. Introduction and objectives

Marine biogeochemistry helps regulate the Earth's climate and controls how the ocean supports life. Climate change, acidification, and deoxygenation alter these cycles, and this will have significant consequences for marine ecosystems and climate feedbacks. Being able to foresee these environmental changes can help us mitigate their consequences [1].

Our approach proposes a deconstruction of the marine biogeochemical system into its main sources of forecastability. We propose an alternative to the "bigger = better" model paradigm by learning these sources of variability with a modular architecture in which each component is treated independently. Because many in-situ biogeochemical observations are collected as depth-resolved profiles [2], we probe the limits of local predictability by isolating what can be forecasted from the water column structure alone. These predictions can eventually be evaluated directly against profile observations, and building on this baseline, we progressively incorporate additional sources of predictability from physics and horizontal interactions.

2. Methods and preliminary results

We train and evaluate on global, gridded ocean data from the Copernicus Marine Service, using monthly-averaged fields from the Global Ocean Biogeochemistry Analysis and Forecast product [3] (1/4° resolution, 2010-2022) together with the GLORYS12V1 Global Ocean Physics Reanalysis [4] (downgraded to 1/4° resolution, 2010-2022) to provide physical context. Because the dominant signal corresponds to seasonality, which masks the more subtle variability, we compute a short-term climatology (2010–2019) to remove the monthly mean value for each variable at each location, training on the residual anomalies instead.

We first compress the water column with a 1D convolutional autoencoder that operates on the depth profiles, which despite their apparent complexity (24 depth levels $\times$ 31 variables) were found to occupy a surprisingly low-dimensional manifold. From as few as 5 latent dimensions, we achieve an accurate reconstruction of the original data (R^2=0.96, rising to 0.98 with 16 dimensions). Across geographic locations, clustering of these embeddings showed coherent patterns consistent with well-known biogeochemical provinces [5], confirming physically meaningful structure. This strong compressibility

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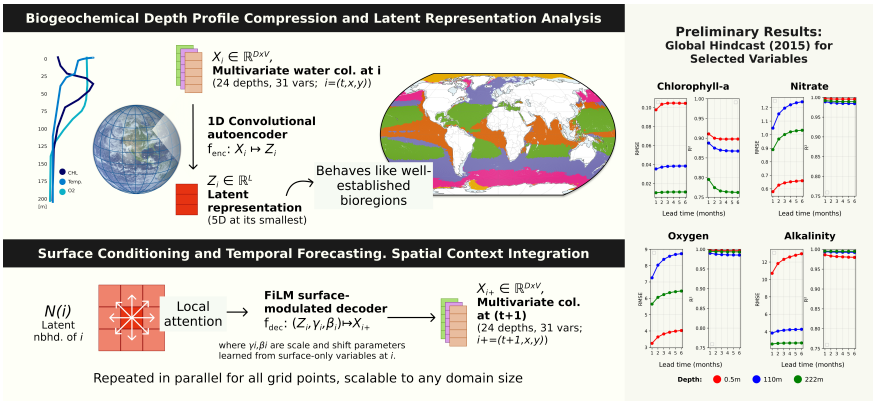

Figure 1. *Visual abstract.*

suggests that the signature, depth-dependent structure of marine biogeochemistry emerges from a small number of underlying processes.

Biogeochemical dynamics are strongly influenced by physical processes, and while we include temperature and salinity in the water column, many observations and physical drivers are surface-bound. We therefore encode additional, surface-level variables into a conditioning vector that is used to inform the column decoder via FiLM (feature-wise linear modulation) [6].

Having established this representation of the column structure, we extend this framework to temporal forecasting. In the simplest formulation, we repurpose the autoencoder to decode the next time step, demonstrating that the learned vertical representation can also capture forecastable dynamics. We also explore a sequence-to-sequence variant. We establish a baseline that achieves competitive forecast skill ($R^2 > 0.91$ and an anomaly correlation coefficient, or ACC > 0.50) by only focusing on the column structure and that can be rolled out autoregressively to assess degradation with lead time.

In preliminary experiments, we augment this baseline with a lightweight latent-neighbourhood module where nearby columns are encoded into a local patch in the latent space, and an attention-based aggregator is used to refine the central representation before prediction. Initial qualitative examination of results suggests improved long-range performance while maintaining grid-independence and modularity, motivating future work with richer spatial context and longer-range dependencies. Looking ahead, a promising direction is to augment the grid-indexed latents with a continuous, coordinate-conditioned encoding [7], making grid-independence a property of the representation itself.

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